

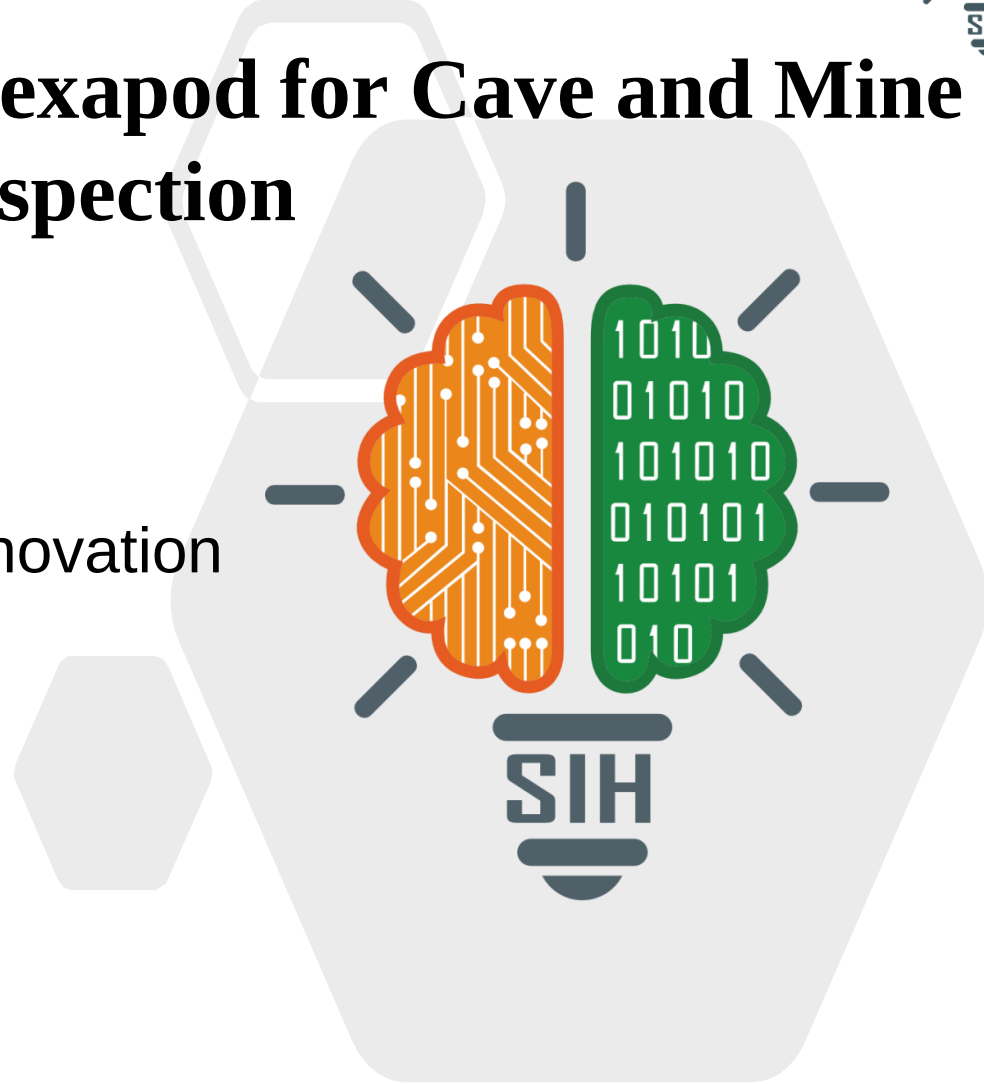
SMART INDIA HACKATHON 2024



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Safety Assessment Hexapod for Cave and Mine Inspection

- **Problem Statement ID** –SIH1593
- **Problem Statement Title**-Student Innovation
- **Theme**-Robotics And Drones
- **PS Category**- Software
- **Team ID**-SVCESIH24HW018
- **Team Name** -Sixfold synergy



IDEA THEME

❖ Proposed Solution

INCIDENT:

A **recent gas-related explosion** occurred on September 21, 2024, at a **coal mine** in Tabas, South Khorasan Province, Iran, resulting in the deaths of at least 51 people and injuring 20 others. The incident was caused by a **methane gas leak** that led to an **explosion while 69 workers** were **inside** the mine.

PROBLEMS :

- Inadequate gas monitoring was a key factor in the incident.
- And a system has to be developed in situations where there is no Wifi or internet connectivity.

SOLUTION :

- Our solution is to develop a hexapod robot equipped with necessary sensors that can move easily over rough and uneven cave surfaces.
- Our hexapod, equipped with dedicated gas sensors and REES52 Raspberry-pi Night Vision camera, detects toxic gases in mines and streams real-time data. It sends alerts when gas levels become dangerous, preventing human exposure.
- This hexapod enhances safety by reducing the need for people to enter hazardous areas.
- The hexapod is designed with advanced mobility features, allowing it to traverse uneven and rugged terrains typically found in caves and mines.
- Its modular architecture enables easy integration of additional sensors or components for enhanced functionality and adaptability in various inspection scenarios.



Hardware Required:

- Raspberry Pi Board 4
- REES52 Raspberry-pi Night Vision camera
- Arduino Mega 2560
- LIDAR sensor
- Multiple Gas detection sensor.
- Ultrasonic Sensors
- 11.1V 5200mAh 3S 35CLiPo battery
- PS2 wireless controller and receiver.
- Buck Boost converter
- 5V Voltage regulator
- 3D printed Hexapod skeleton made up of PLA
- 18 towerpro metallic 95 mg995 servo motors
- Zigbee

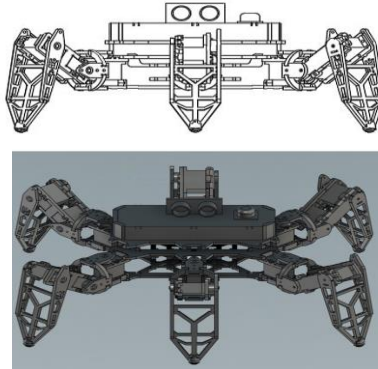
Software Required:

- Blender(3D modelling)
- Arduino IDE
- Pytorch
- HTML,CSS,JS(Web Page)

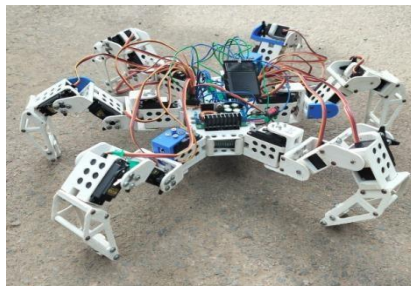
Tap to view STL file :

https://drive.google.com/drive/folders/1rz2qqRaDbn7_jmnggyCEwfnA5I6zGEp7

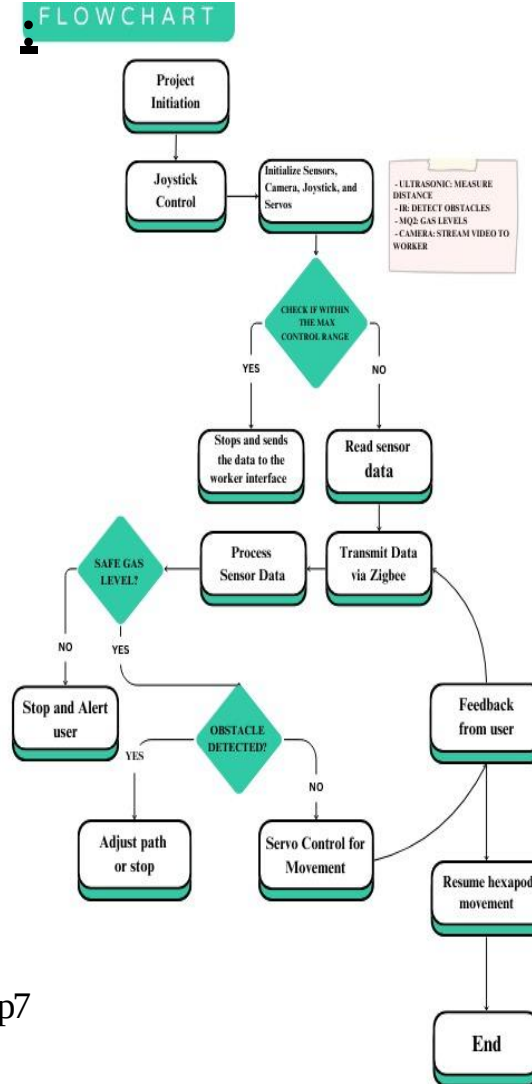
3D Rendering



Hexapod Structure

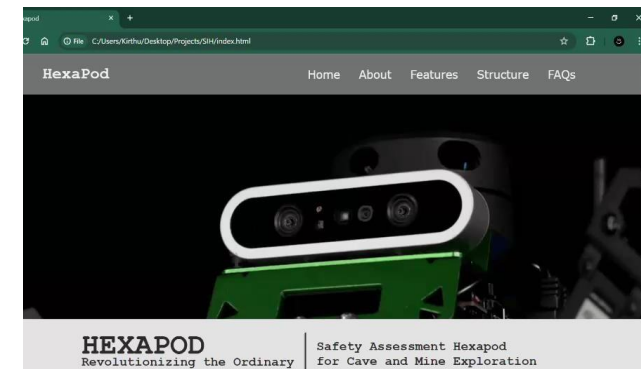


Workflow and Flowchart:

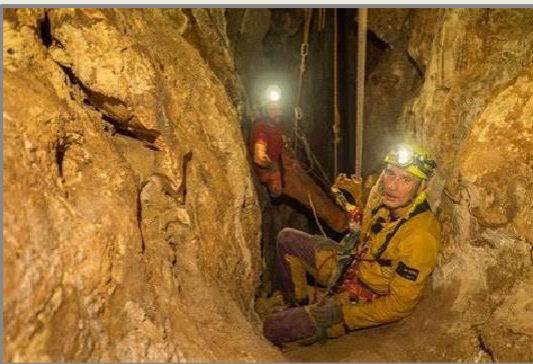


1. The hexapod is controlled via a joystick to monitor the environment using sensors.
2. Real-time sensor data and video feed are sent to workers through a user interface via Zigbee
3. If harmful gases are detected, the workers can visualize the alert and relevant data on the interface, enabling them to take immediate safety measures.
4. If no harmful gases are detected, the workers can safely approach the hexapod's current position, then send it forward for further exploration
5. This process repeats until the hexapod reaches its goal, ensuring safe and informed movement.

Our webpage:



Potential impact on the target audience:



- Reduces risks to coal miners and improves navigation in hazardous mines.
- Enhances safety and efficiency by navigating rugged surfaces and providing real-time data
- Sets new standards and offers valuable insights for future exploration missions.

Benefits of the solution:

- **Stable navigation on rocky surfaces.**
- Since the mines may not contain wifi access we are using zigbee for data communication.
- **Real-Time Data Transmission:** Improved coordination and decision-making in order to **prevent fatal tragedies.**
- **Reduced Risk:** Minimizes danger to **miners and cave explorers.**
- **Faster Operations:**Speeds up exploration and mapping of cave systems and mines.

Analysis of the feasibility of the idea

- LIDAR sensors for 3D mapping, night vision cameras for visuals, and a dedicated array of sensors for air quality, ensuring safe navigation in mines and caves.
- Zigbee modules ensure efficient sensor data transmission to mine workers without needing cellular networks or Wi-Fi, perfect for mines and caves.
- A LiPo battery provides lightweight, high-energy, fast-recharge power, perfect for a compact hexapod design with extended operation and easy replacement.

Potential challenges and risks

- Communication and real time data transmission from mine is a challenge since there wasn't network availability inside the mine or cave .
- Irregular surfaces could make the hexapod navigation difficult.
- Transmission of Video live feed
- Sparks generated from switches or damaged battery could be dangerous

Strategies for overcoming these challenges

- We will be using **two Zigbee** modules: one will be mounted **on the hexapod**, and the other will be on the **joystick with display**, operated by the miner.
- Dedicated night vision cameras are used along with **LIDAR AND ML controlled path planning algorithms** to overcome the navigation difficulty.
- **Usage of Wifi 6 Technology in mines is effective and could help with transmission of video**
- Proper insulation of switches and battery would protect the sparks or damage of battery.

References:

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